BUSINESS INTELLIGENCE AND DATA ANALYTICS

Institute of Analytics
Recognized Programme

University of Petra

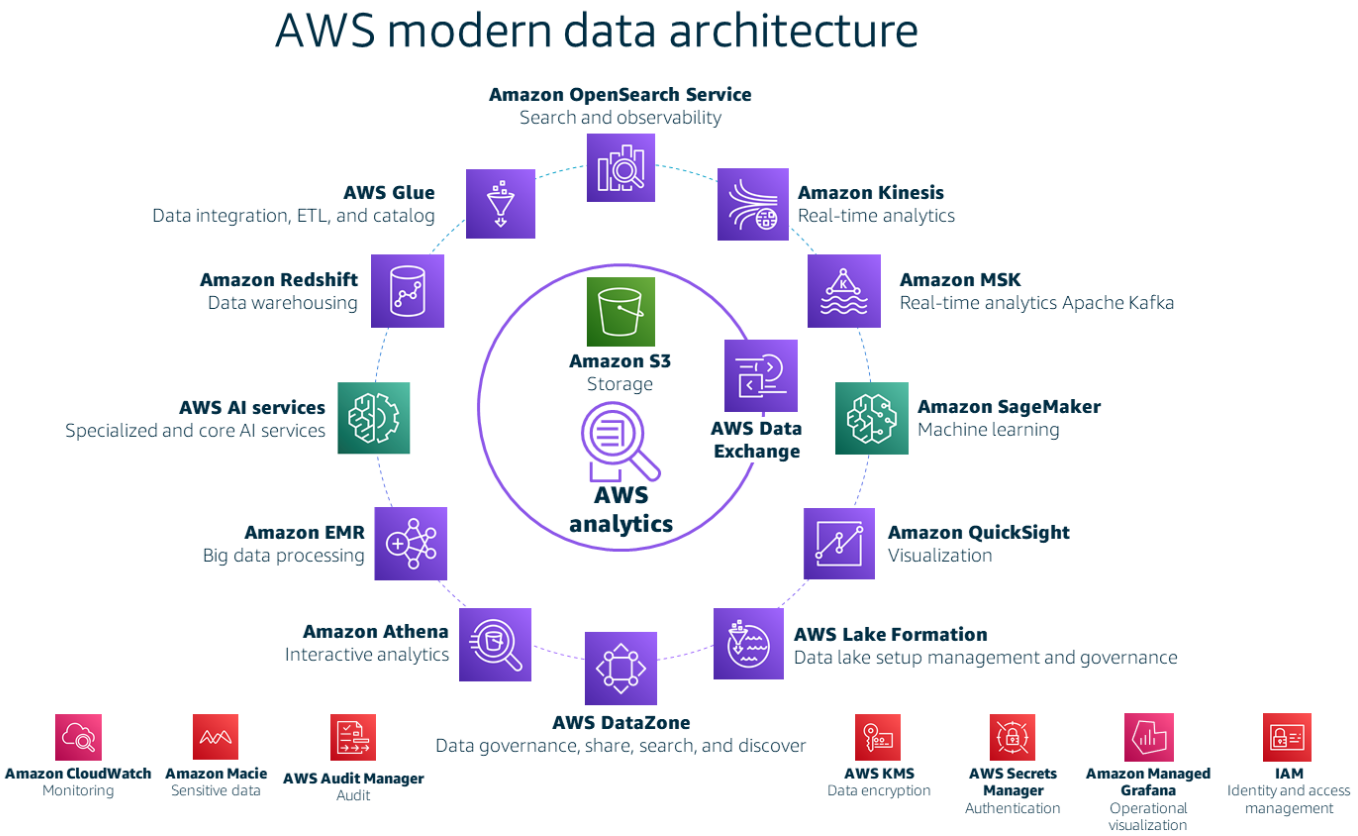
307401 Big Data

Introduction to AWS Data Engineering

2024-2

Overview of AWS Data Engineering

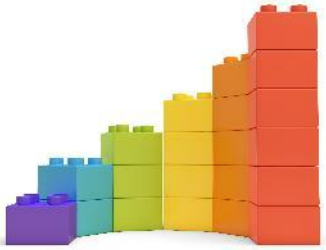
- **Data Engineering**: Design and build systems for collecting, storing, and analyzing data.
- **Purpose**: Transform raw data into actionable insights using robust, scalable, and secure data pipelines.
- **Core AWS Services**:
  - **S3**: Scalable object storage for raw/staged/curated data.
  - **Glue**: Serverless data integration (ETL), data cataloging.
  - **Athena**: Serverless querying on S3 using SQL.
  - **Redshift**: Data warehousing and analytics.
  - **Kinesis**: Real-time data streaming.
  - **Lambda**: Serverless compute for lightweight ETL.
  - **EMR**: Managed Hadoop/Spark clusters.
  - **Data Pipeline / Step Functions**: Workflow orchestration.



A data pipeline in its simplest terms

- The data pipeline is the infrastructure that you build to support data-driven decision-making.
- At the most basic level, any data pipeline infrastructure must be able to (1) bring data in, (2) store it, and (3) provide the means to work with the data to derive insights.
- As the data engineer or data scientist, it's your job to build the pipeline to figure out what's appropriate and how you're going to do it.
- Often this will include exploring and experimenting to get to know the data and the best way to derive value from it.

Collect data



Store and process data

Build something useful with data

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Work backwards to design your infrastructure

- The number of variations and options to create a data pipeline are broad.
- The key to designing an effective decision-making infrastructure is to start with the business problem to be solved or decision to be made.
- Then, build the pipeline that best suits that use case.
- The cloud makes it easier to start up different infrastructure to serve different use cases.
- An organization doesn't need to invest in an infrastructure that tries to serve all business needs with a single database, server, or technology stack.
- For each use case, start with the end in mind, and build the data pipeline that supports it.

2. What data do you need to support this?

1. What decision are you trying to make?

Weigh the trade-offs of cost, speed, and accuracy.

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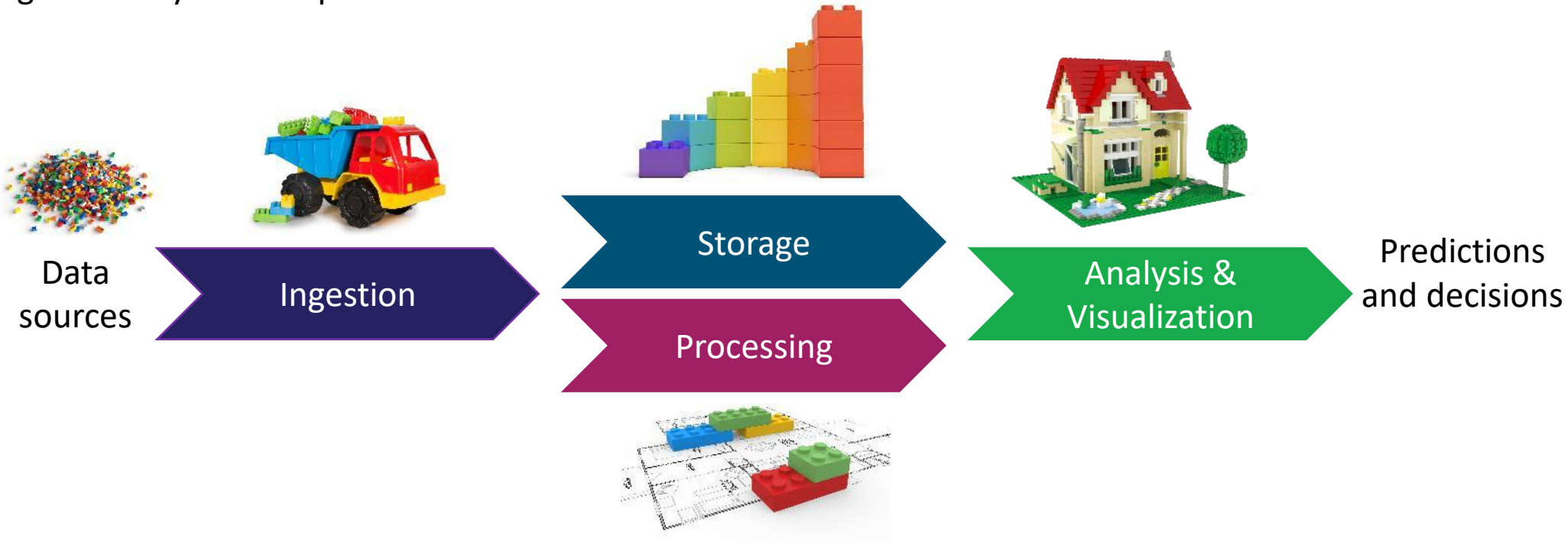
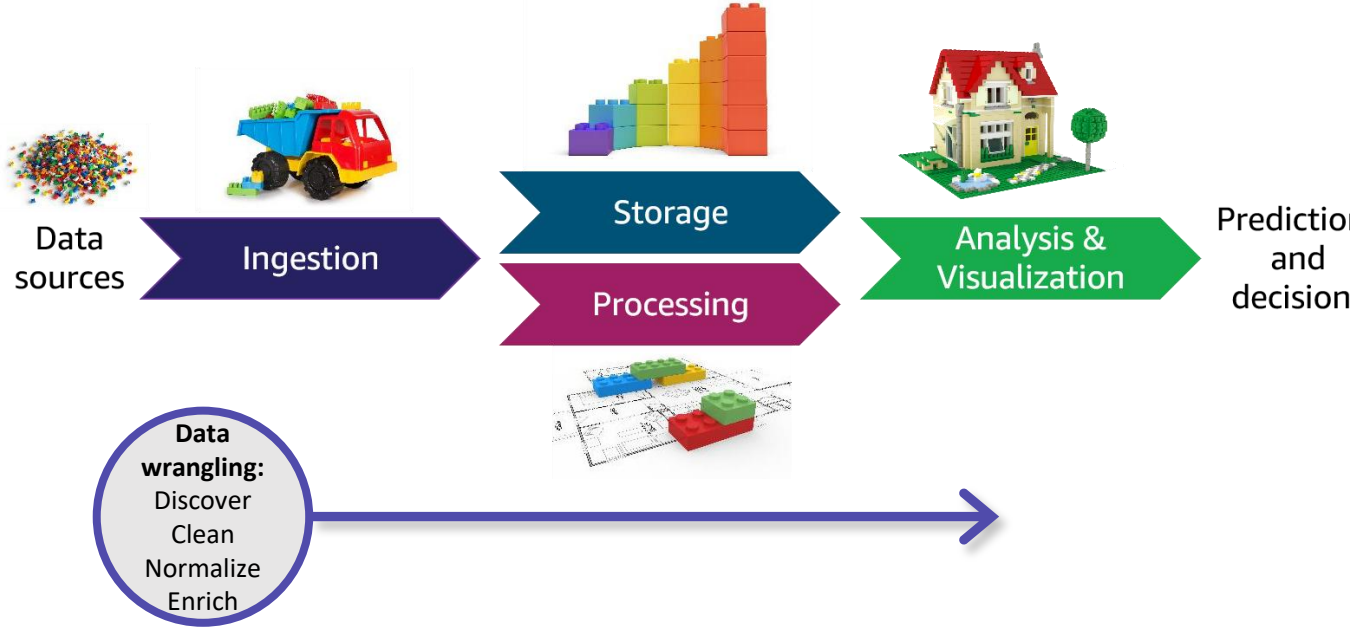
Layers of the pipeline infrastructure

- The infrastructure layers that you need to build include methods to ingest and store data from the data sources that you have identified.
- You need to make the stored data accessible for decision-making processes.
- You must also create the infrastructure to process, analyze, and visualize data by using tools that are appropriate to the use case.
- To build an appropriate infrastructure, you will need to understand the nature of the data and the intended type of processing and analysis to be performed.



Actions taken with data in the pipeline

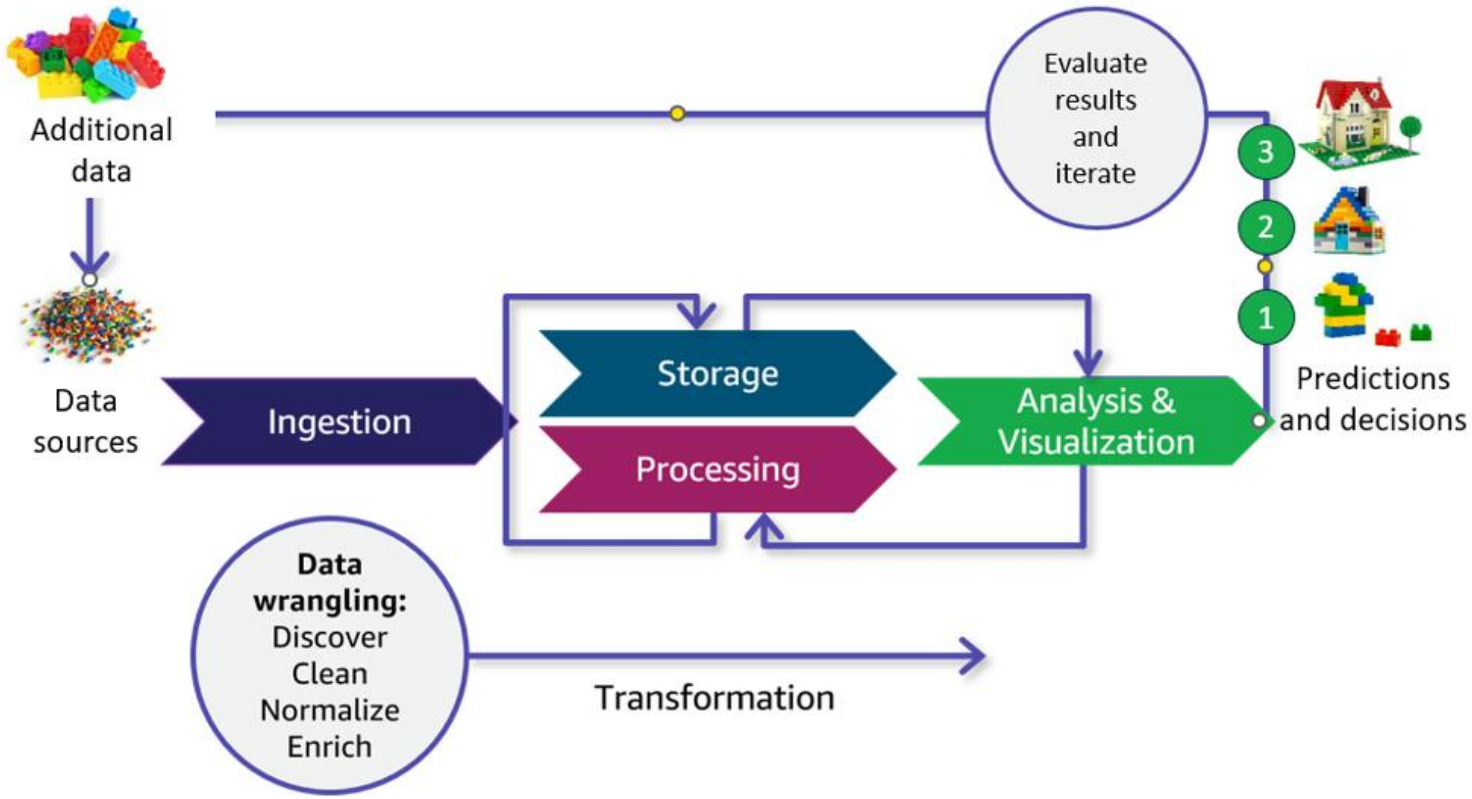
- The pipeline must support the ability to discover details about the data and how it fits (or doesn't fit) your intended outcomes.
- You might start with a hypothesis and use BI tools or other mechanisms to discover more information about the data, and then experiment and see where it takes you.
- Data will almost always be transformed as it moves through the pipeline. This might include modifying the format to support a specific analysis tool or replacing values (for example, zeroes in place of nulls).
- *Data wrangling* is a term that is used to describe the ways in which data is manipulated and transformed from its raw state into more meaningful states that downstream processes or users can use.



Transformation

Iterative processing through the pipeline

- Another key characteristic of deriving insights by using your data pipeline is that the process will almost always be iterative.
- For example, in this illustration, the initial iteration (number 1) yielded a result that wasn't as defined as was desired. Therefore, the data scientist refined the model and reprocessed the data to get a better result (number 2).
- After reviewing those results, they determined that additional data could improve the detail available in their result, so an additional data source was tapped and ingested through the pipeline to produce the desired result (number 3).



Common data pipeline questions

Data engineer

- Does the organization have the data that addresses the need? Where is the data stored and in what format?
- Will I need to combine data from multiple sources?
- What is the type and quality of data? What will be the source of truth?
- What are the security requirements for this data? Who needs access to it and in what state?
- What types of mechanisms are needed to transfer the data from its locations into the pipeline?
- How much data is there, and how frequently is it updated or processed?
- How important is speed when data is requested?

- What can the data tell me?
- How will I evaluate the results?
- What kind of visualization do I need?
- Which formats and tools are the analysts familiar with?
- Do I need a big data framework?
- Which type of AI/ML models fit?
- What is the simplest way to implement AI/ML?

Data scientist

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Key takeaways: The data pipeline – infrastructure for data-driven decisions

- A data pipeline provides the infrastructure for data-driven decision-making.
- When designing a pipeline, start with the business problem to be solved and work backwards to the data.
- The pipeline includes layers to ingest, store, process, and analyze and visualize data passing through it.
- Data wrangling refers to how data is acted upon as it passes through the pipeline. Tasks include discovery, cleaning, normalization, transformation, and augmentation.
- Data is processed iteratively to evaluate and improve upon results.

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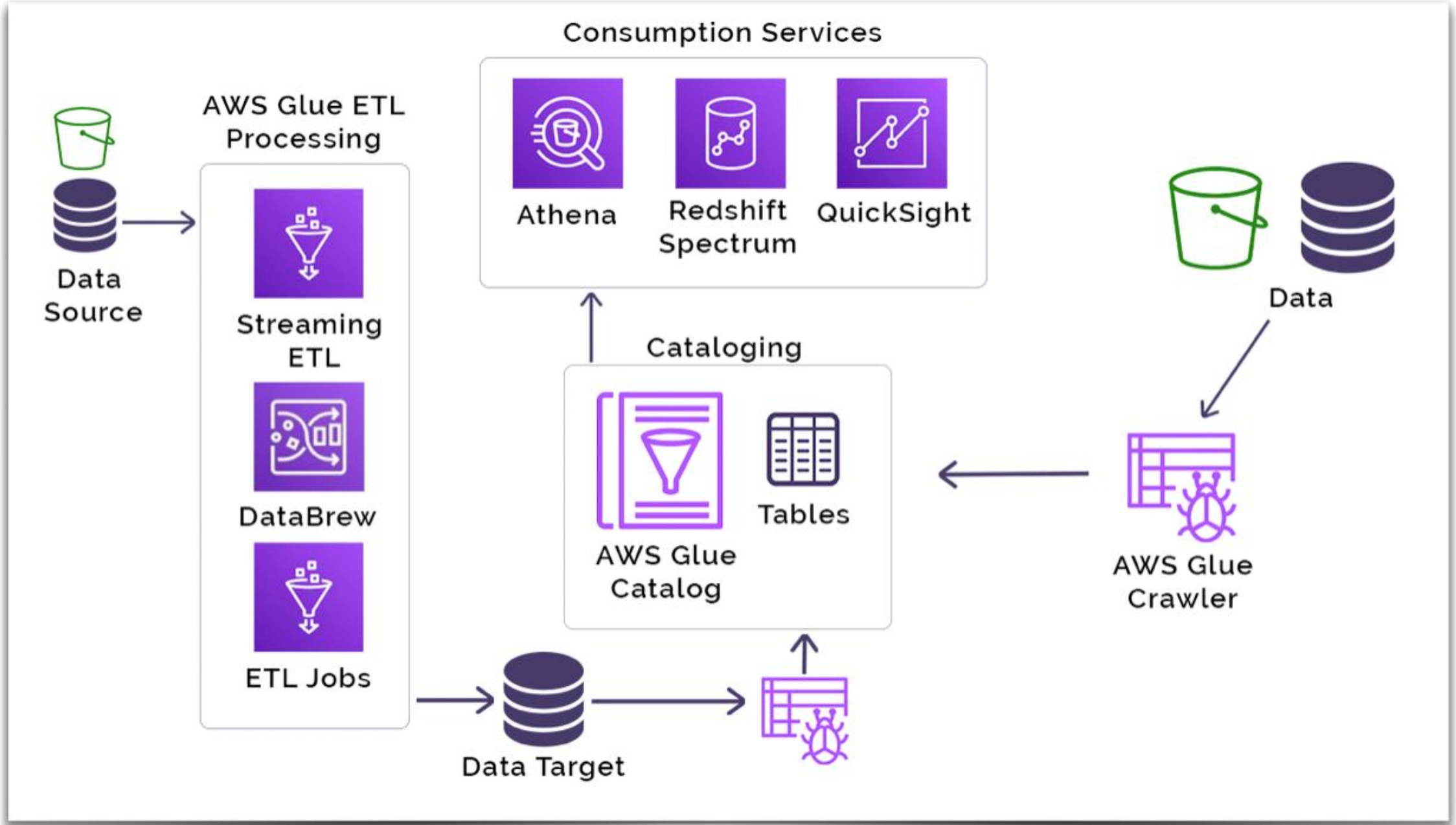
Amazon Glue


Consumption Services
Athena
Redshift Spectrum
QuickSight
AWS Glue ETL Processing
Data Source
Streaming ETL
DataBrew
ETL Jobs
Cataloging
AWS Glue Catalog
Tables
Data
AWS Glue Crawler
Data Target

Introduction to AWS Glue

What is AWS Glue?

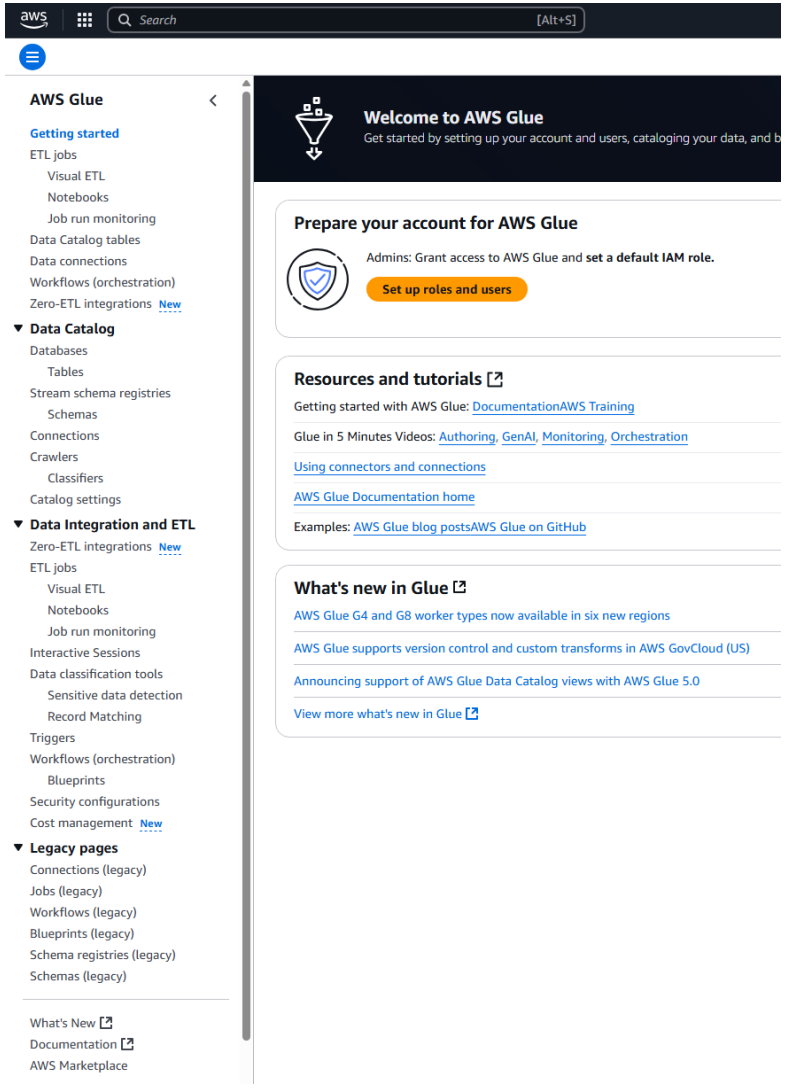
- AWS Glue is a fully managed, serverless data integration service that simplifies the process of discovering, preparing, and combining data for analytics, machine learning, and application development.

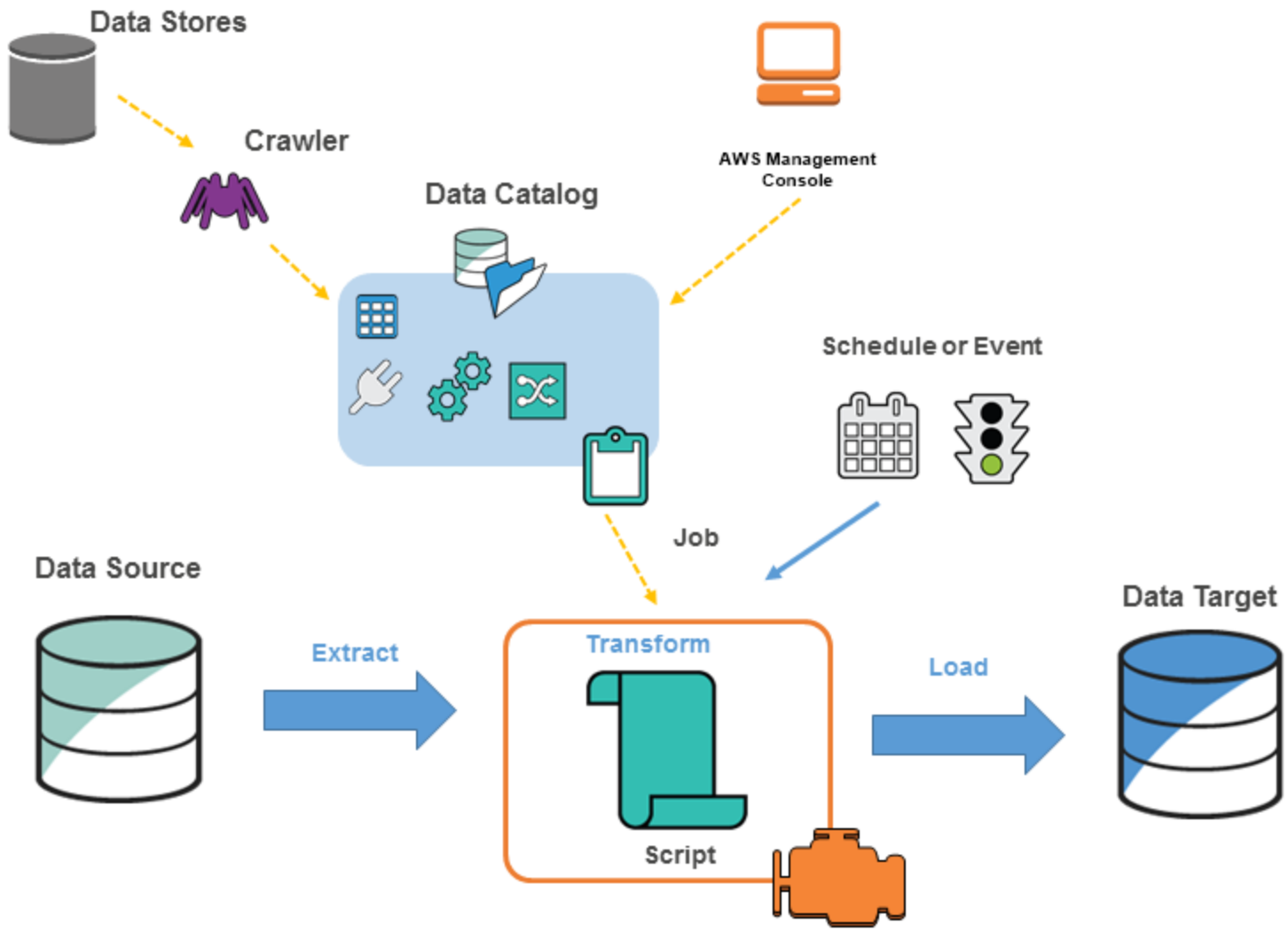
Key Features:

- **Serverless Architecture:** No infrastructure to manage; AWS handles provisioning and scaling.
- **Data Cataloging:** Centralized metadata repository for all data assets.
- **Automated ETL:** Automatically generates code to extract, transform, and load data.
- **Support for Multiple Data Sources:** Integrates with various AWS and third-party data sources.

Core Components of AWS Glue

1. **AWS Glue Data Catalog:** A centralized metadata repository that stores table definitions, job metadata, and other control information to manage your ETL environment.
2. **Crawlers:** Automated processes that scan data sources to infer schemas and populate the Data Catalog.
3. **Jobs:** Scripts that define the ETL process, written in Python or Scala, which can be generated automatically or customized.
4. **Triggers:** Mechanisms to initiate jobs based on schedules or events.
5. **Workflows:** Orchestrate multiple jobs and crawlers to build complex ETL pipelines.
6. **Transformers**: Built-in and custom data transformation functions.




Data Stores
Crawler
Data Catalog
AWS Management Console
Schedule or Event
Job
Data Source
Extract
Transform
Script
Load
Data Target

AWS Glue Data Catalog

Purpose:

- Acts as a persistent metadata store for all your data assets, enabling data discovery and schema management.

Features:

- **Schema Versioning:** Tracks changes to data schemas over time.
- **Integration:** Works seamlessly with Amazon Athena, Redshift Spectrum, and Amazon EMR.
- **Security:** Supports fine-grained access control using AWS Identity and Access Management (IAM).

AWS Glue Crawlers

Functionality:

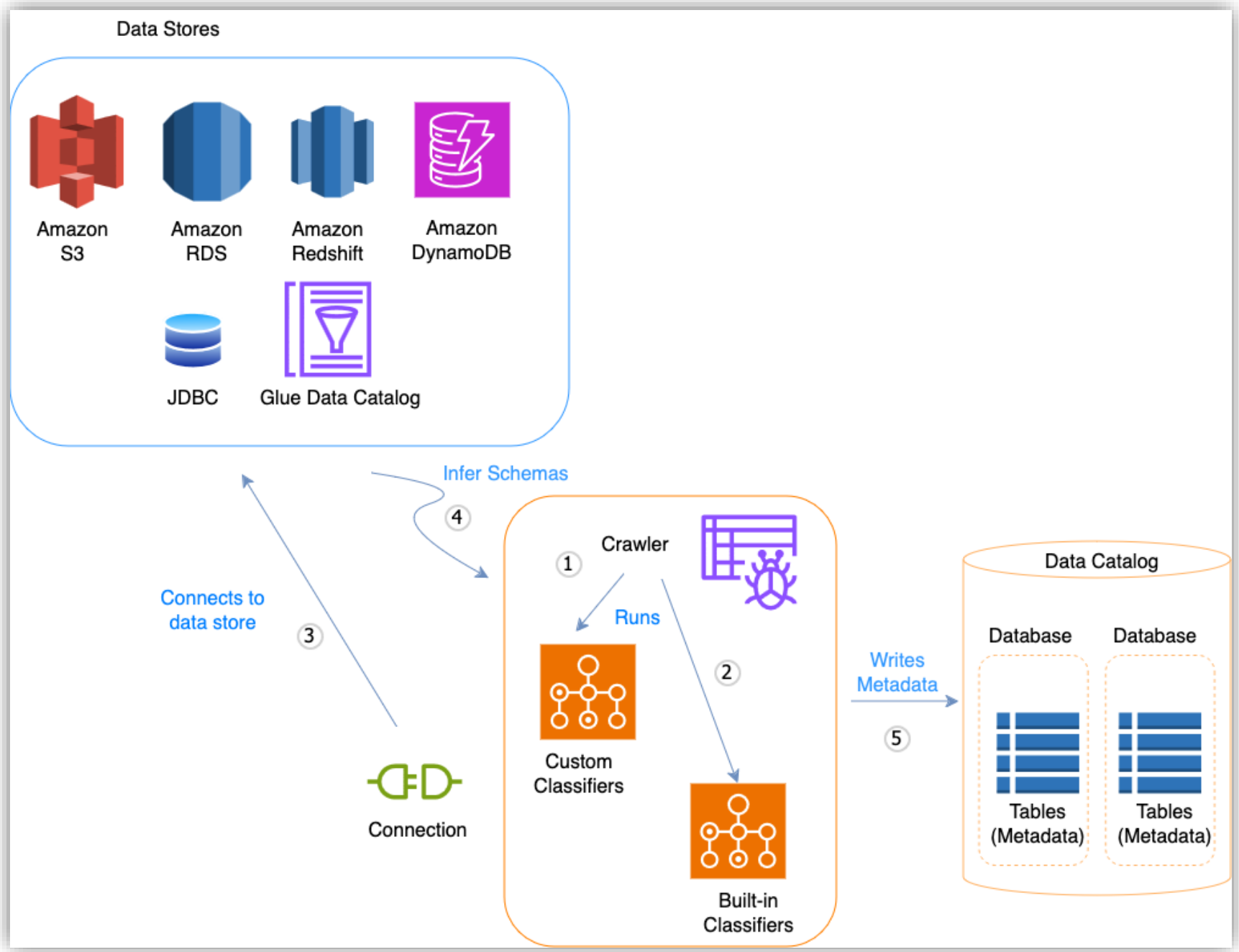
- Connect to various data sources, infer schemas, and populate the Data Catalog with metadata.

Supported Data Stores:

- Amazon S3, Amazon RDS, Amazon Redshift, DynamoDB, and more.

Custom Classifiers:

- Define custom logic to classify data formats not supported by default.


Data Stores
Amazon S3
Amazon RDS
Amazon Redshift
Amazon DynamoDB
JDBC
Glue Data Catalog
Infer Schemas
4
Crawler
1
Runs
2
Connects to data store
3
Connection
Custom Classifiers
Built-in Classifiers
Writes Metadata
5
Data Catalog
Database
Database
Tables (Metadata)
Tables (Metadata)

AWS Glue Jobs

Types of Jobs:

- **ETL Jobs:** Perform extract, transform, and load operations.
- **Streaming Jobs:** Process real-time data streams using Apache Spark Structured Streaming.

Development Options:

- **AWS Glue Studio:** Visual interface to create and manage jobs.
- **Script Editor:** Write custom scripts in Python or Scala.

Job Monitoring:

- Track job runs, monitor logs, and set up alerts for failures or delays.

Triggers and Workflows

Triggers:

- **Scheduled Triggers:** Initiate jobs at specified times.
- **Event-Based Triggers:** Start jobs in response to events like new data arrival.

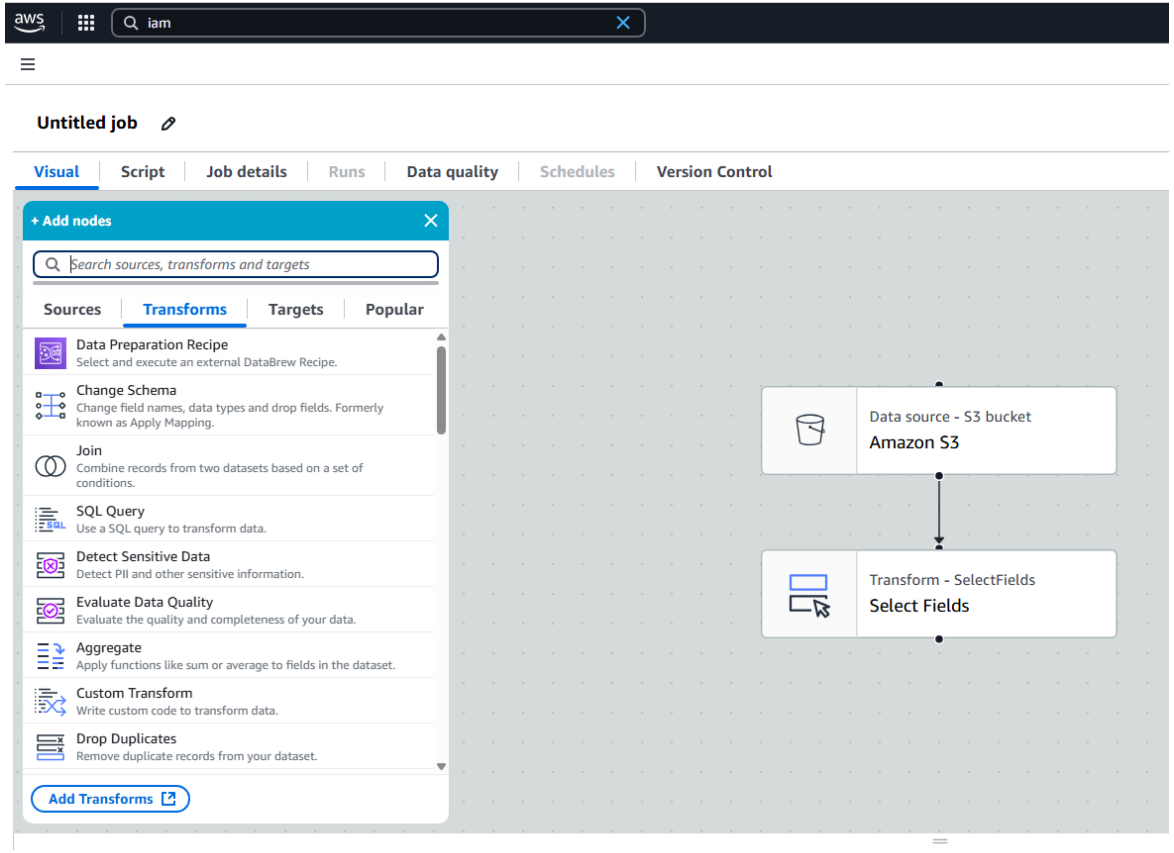
Workflows:

- Define a sequence of jobs and crawlers with dependencies to automate complex ETL processes.

AWS Glue Studio

Features:

- User-friendly interface to create, run, and monitor ETL jobs without writing code.
- Drag-and-drop functionality to design data flows.
- Automatic code generation with the option to customize scripts.



AWS Glue DataBrew

Overview:

- Visual data preparation tool for data analysts and scientists.

Capabilities:

- Clean and normalize data without writing code.
- Apply transformations, handle missing values, and create data profiles.

Integration with Other AWS Services

Amazon S3:

- Store and retrieve data for processing.

Amazon Redshift:

- Load transformed data into Redshift for analytics.

Amazon Athena:

- Query data directly from S3 using SQL.

Amazon EMR:

- Process large datasets using big data frameworks like Apache Spark and Hadoop.

Security and Compliance

Data Encryption:

- Supports encryption at rest and in transit.

Access Control:

- Fine-grained permissions using IAM roles and policies.

Audit Logging:

- Integration with AWS CloudTrail for monitoring API calls and user activities.

amazon ATHENA

Introduction to Amazon Athena

- **What is Athena?**
  - A **serverless** query service for **directly analyzing data in Amazon S3** using standard SQL.
  - **No infrastructure** to set up or manage — you pay only for the data scanned per query.
- **Trino and Athena**
  - Athena is built on the open-source Trino query engine
  - Trino (formerly PrestoSQL) enables fast, scalable query processing
  - Distributed SQL query engine for big data processing
- **Main Advantages:**
  - **Ad hoc querying**: instantly run queries without ETL pipelines.
  - Data exploration and discovery
  - Serverless analytics
  - **Support for various formats**: CSV, JSON, ORC, Parquet, Avro.
  - **Integration with AWS Glue** for metadata management.
  - **Security**: IAM controls query and data access.

How Athena Works

Architecture Overview:

- Data is stored in **Amazon S3**.
- Metadata is stored in **AWS Glue Data Catalog** or internal Athena catalog.
- SQL engine processes queries via a distributed serverless backend (based on Presto).

Query Flow:

1. Define table schema (pointing to S3 location).
2. Submit SQL query via Athena console, SDK, or CLI.
3. Query results are returned; metadata is stored.

Core Components of Amazon Athena

1. Data Sources:

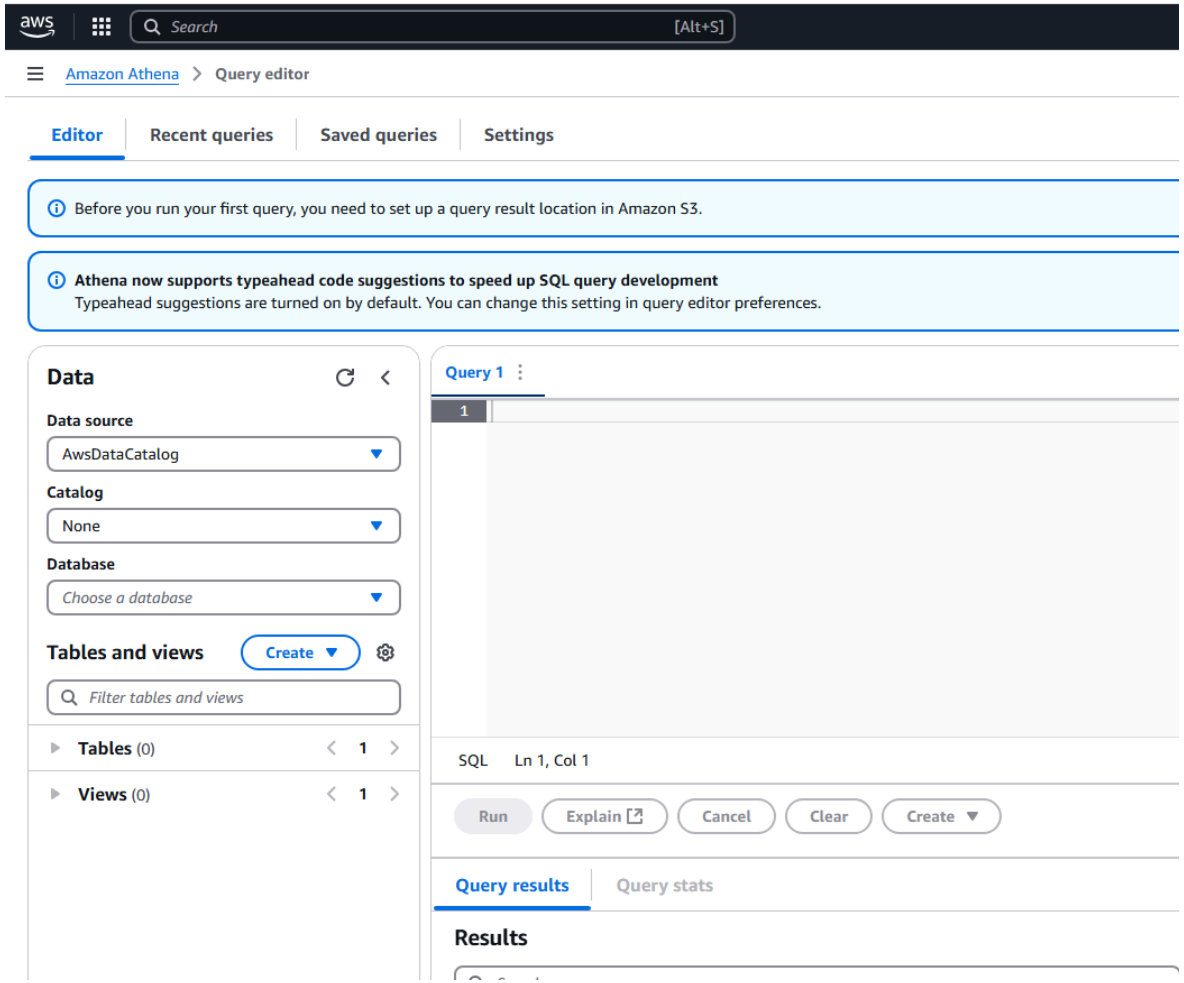
- Primarily queries data stored in Amazon S3.
- Supports various data formats, including CSV, JSON, ORC, Avro, and Parquet.

2. Query Engine:

- Built on Presto, an open-source distributed SQL query engine optimized for low-latency data analysis.

3. Integration with AWS Glue:

- Utilizes AWS Glue Data Catalog for managing metadata and schema definitions.



Athena and S3 Integration

Data Source:

- Athena **only reads data** from **Amazon S3**.

Common File Formats:

- CSV, JSON, ORC, Parquet, Avro, and more.

Performance Considerations:

- Columnar formats like Parquet and ORC reduce data scanned.
- Partitioning directories in S3 improves performance.

Cost Optimization:

- Compress and partition data to reduce scan size.

AWS Glue Data Catalog Integration

Glue's Role:

- AWS Glue crawls and catalogs data in S3
- Makes data discoverable by Athena for querying
- Acts as a **central metadata repository** for Athena.

Athena + Glue Integration:

- Athena can use Glue tables, databases, and crawlers.
- Enables table reuse and consistency across AWS services.

Glue Crawlers:

- Automatically detect schema from S3 data and create metadata tables.

Schema Evolution in Athena

- Athena supports schema-on-read approach
- Dynamically infers structure during query execution
- Handles schema changes without explicit alterations
- New field addition doesn't break existing queries
- AWS Glue Crawler updates metadata after schema changes

- SCHEMA EVOLUTION EXAMPLE
  - Assume you have an Athena Table defined
  - When a new field is added to the source data:
    - Queries will still work with fields in original definition
    - To access new field data, table DDL must be updated

Partitioning in Athena

- Organizes data based on specific column values
- Creates subdirectories in S3 for each partition value
- Enhances query performance by skipping irrelevant data
- Best Practice: Partition by date for time-series data

CHOOSING PARTITION KEYS

- Pick keys that support common query filters
- Use columns with relatively low cardinality
- Avoid heavily skewed partition values
- Time-based partitioning works well for growing datasets

OPTIMIZING QUERY PERFORMANCE

1. **Appropriate File Formats:** Use columnar formats (Parquet, ORC)
2. **Partitioning:** Organize data to skip unnecessary scans
3. **Predicate Pushdown:** Use filters in WHERE clause
4. **Bucketing:** Organize data within partitions
5. **Control Data Volume:** Limit time ranges, use sampling

```
s3://your-bucket/your-data/
├── date=20220101
│   ├── file1.csv
│   └── file2.csv
├── date=20220102
│   ├── file3.csv
│   └── file4.csv
└── date=20220103
    ├── file5.csv
    └── file6.csv
```

Predicate Pushdown in AWS Athena

Predicate pushdown is a query optimization technique used in AWS Athena that significantly improves query performance by reducing the amount of data that needs to be scanned and processed.

How Predicate Pushdown Works

1. **Filter Application at Source**: When you include filter conditions in the WHERE clause of your query, these conditions (predicates) are "pushed down" to the data source level.
2. **Early Data Filtering**: Instead of loading all data and then filtering it, the conditions are applied while reading the data, which means only relevant data is loaded.
3. **Metadata Utilization**: For columnar formats like Parquet and ORC, Athena uses file metadata (min/max values per column in each file block) to determine which data blocks can be skipped entirely.

Example

```
SELECT customer_name, order_total FROM orders WHERE order_date > '2023-01-01' AND region = 'EMEA';
```

With predicate pushdown:

- Athena can skip reading any file blocks where the min/max values for 'order_date' indicate all values are before 2023-01-01
- Similarly, it can skip blocks where 'region' does not contain 'EMEA'

Benefits

- **Reduced Data Scanning**: Only relevant data blocks are read
- **Lower Costs**: AWS Athena charges based on amount of data scanned
- **Faster Queries**: Less data to process means faster execution time
- **Improved Scalability**: More efficient resource utilization

Best Practices

- Use file formats that support predicate pushdown (Parquet, ORC)
- Structure your WHERE clauses to filter as early as possible in the query
- Sort data on frequently filtered columns when writing to S3
- Partition data to complement predicate pushdown for compound performance gains
- Predicate pushdown is one of the most effective techniques for optimizing Athena queries, especially when working with large datasets.

Optimizing Queries with Compression, Buckets, and Partitions

Buckets

- **Definition:** Logical grouping of related records based on **high-cardinality** fields.
- **Purpose:** Speed up query performance by minimizing full dataset scans.
- **Example:** Splitting taxi trips into separate buckets by each day's timestamp.

Compression

- **Purpose:** Store data using compressed formats (e.g., GZIP, Parquet) to:
  - Save S3 storage space.
  - Reduce the amount of data Athena scans (saving cost).

Partitioning

- **Definition:** Splitting data into logical directories based on **low-cardinality** fields.
- **Example:** Partitioning taxi trips by **payment type** (Credit card, Cash).

"Athena's cost is based on data scanned — so optimizing file formats and partitions can massively reduce costs in Big Data environments."

Optimizing Queries with Compression, Buckets, and Partitions

Technique	Best for	Example	Benefit
Compression	Any large dataset	Compress CSV to GZIP	Lower storage and scan costs
Bucketing	High cardinality fields	Group by pickup timestamp	Speed up random queries
Partitioning	Low cardinality fields	Split by payment type	Scan only relevant partitions

Golden Rule: Less data scanned = Faster query + Lower cost

Note: Buckets in compression is different than S3 storage buckets

Term	Meaning	Context
S3 Buckets	Storage containers in Amazon S3 to hold files (objects).	Cloud storage. Think: folders for storing any type of data (CSV, images, backups).
Bucketing (in Athena/Glue/Presto SQL)	A **data organization technique**: splitting data into logical groups ("buckets") based on the values of a column.	Database optimization. Think: dividing a huge table into smaller "mini-tables" inside a database to speed up queries.

Handling Large Files

- Use appropriate file formats (Parquet/ORC)
- Implement effective partitioning strategies
- Apply bucketing for high-cardinality columns
- Control query scope with filters
- Use columnar compression

Create a Table with Partition in Athena

```
CREATE EXTERNAL TABLE sales_records ( product_id INT, sales_amount DECIMAL(10, 2), transaction_date DATE )
PARTITIONED BY (year INT, month INT)
LOCATION 's3://your-bucket/sales-data/';
```

Module 4 Lab: Querying Data Using Amazon Athena

In this lab, you will learn to query CSV data stored in Amazon S3 using **Amazon Athena** and **AWS Glue**.

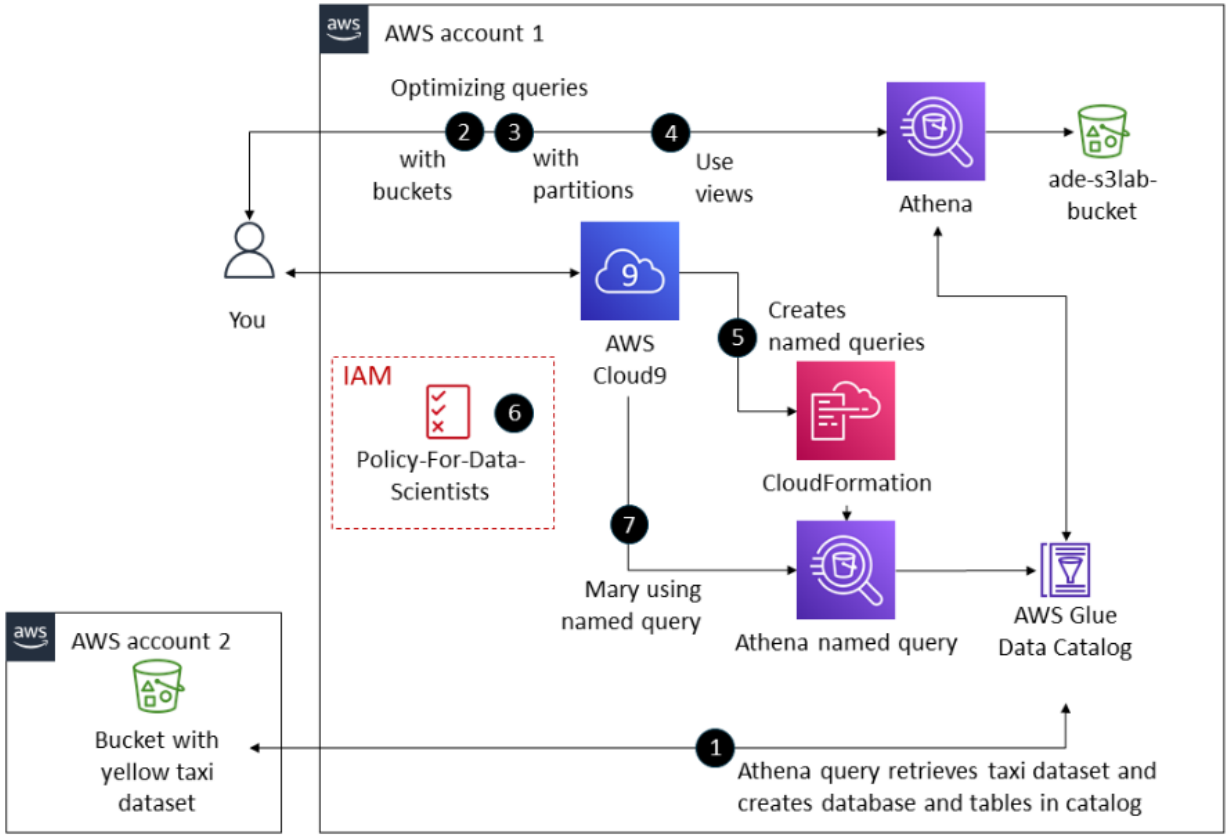
Athena enables SQL queries on S3 data without moving it, while Glue helps define and manage metadata.

You will implement a lab in AWS Academy Data Engineering Course (Module 4) where we simulate the role of a data scientist building a scalable, permission-controlled solution that supports team-wide access following the principle of least privilege.

Objectives

You will:

- Create a Glue database and table via the Athena editor
- Define schema and use bulk column features
- Query and optimize data in S3 with Athena
- Create views and named queries
- Deploy named queries using CloudFormation
- Validate IAM access through AWS CLI
- This training uses a 2017 taxi trip dataset and is restricted to required AWS services only.



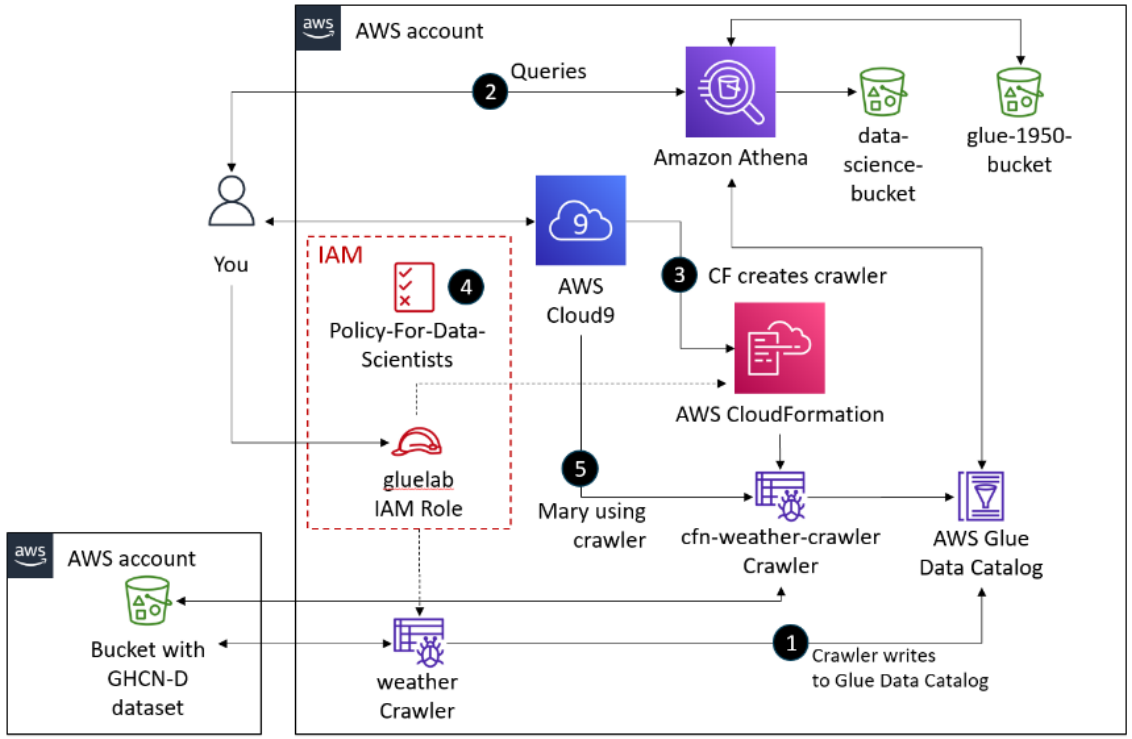
Module 7 Lab: Performing ETL on a Dataset by Using AWS Glue

In this lab, you will learn how to use AWS Glue to import a dataset from Amazon Simple Storage Service (**Amazon S3**).

You will then extract the data, transform its schema, and load the dataset into an **AWS Glue** database for later analysis by using Athena.

After completing this lab, you will be able to do the following:

- Access AWS Glue in the AWS Management Console and create a **crawler**.
- Create an AWS Glue database with tables and a schema by using a crawler.
- Query data in the AWS Glue database by using Athena.
- Create and deploy an AWS Glue crawler by using an AWS CloudFormation template.
- Review an AWS Identity and Access Management (IAM) policy for users to run an AWS Glue crawler and query an AWS Glue database in Athena.
- Confirm that a user with the IAM policy can use the AWS Command Line Interface (AWS CLI) to access the AWS Glue database that the crawler created.
- Confirm that a user can run the AWS Glue crawler when source data changes.



amazon REDSHIFT

Amazon Redshift

Amazon Redshift is a fully managed data warehouse service offered by Amazon Web Services (AWS).

It is designed for large-scale data storage and fast query performance, making it suitable for data analytics and business intelligence workloads.

Here are some key features and benefits of Amazon Redshift:

Key Features and Benefits:

- **High Performance**:
  - Redshift uses columnar storage and data compression to reduce the amount of data read from disk, improving query performance.
  - Massively parallel processing (MPP) architecture allows Redshift to distribute and parallelize queries across multiple nodes.
- **Scalability**:
  - Redshift allows you to start with a small cluster and scale up to petabytes of data.
  - You can easily resize your Redshift cluster by adding or removing nodes.
- **Cost-Effective**:
  - Pay only for the resources you use with on-demand pricing.
  - Reserved instance pricing and the ability to pause and resume clusters can further reduce costs.
- **Managed Service**:
  - AWS manages the setup, operation, and scaling of Redshift, including backups, patching, and monitoring.
  - Automatic backups and snapshots provide data protection and recovery.
- **Data Integration**:
  - Redshift integrates with various AWS services such as Amazon S3, Amazon RDS, Amazon DynamoDB, and AWS Glue for data ingestion and ETL processes.
  - Supports integration with third-party ETL tools and BI platforms.
- **Advanced Analytics**:
  - Supports SQL-based querying and analytics.
  - Redshift Spectrum allows you to run queries on data stored in Amazon S3 without loading it into Redshift.
  - Machine learning integration with Amazon SageMaker for predictive analytics.

Core Components of Amazon Redshift

1. Clusters:

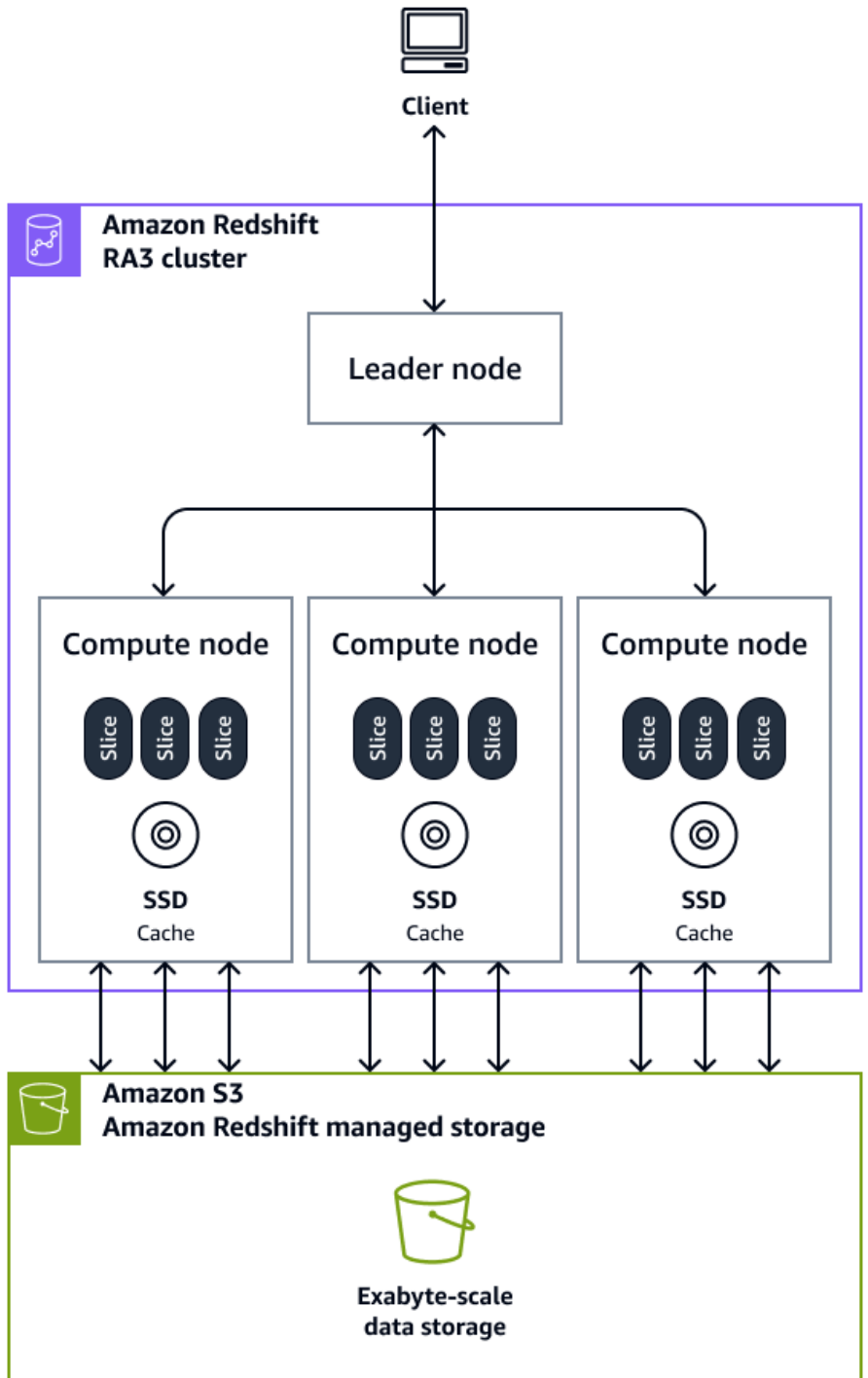
- The core infrastructure component, composed of one or more compute nodes.

2. Leader Node:

- Manages communications with client programs and all communication with compute nodes.

3. Compute Nodes:

- Handle data processing tasks and store data.



Data Storage and Compression

Columnar Storage:

- Data is stored in columns, optimizing query performance and reducing I/O.

Compression:

- Redshift applies compression to reduce storage requirements and improve performance.

sID	product	location	available
1	chair	Boston	15
2	chair	Ohio	6
3	chair	Denver	9

row-oriented

column-oriented

sID	product
1	chair
2	chair
3	chair

sID	location
1	Boston
2	Ohio
3	Denver

sID	available
1	15
2	6
3	9

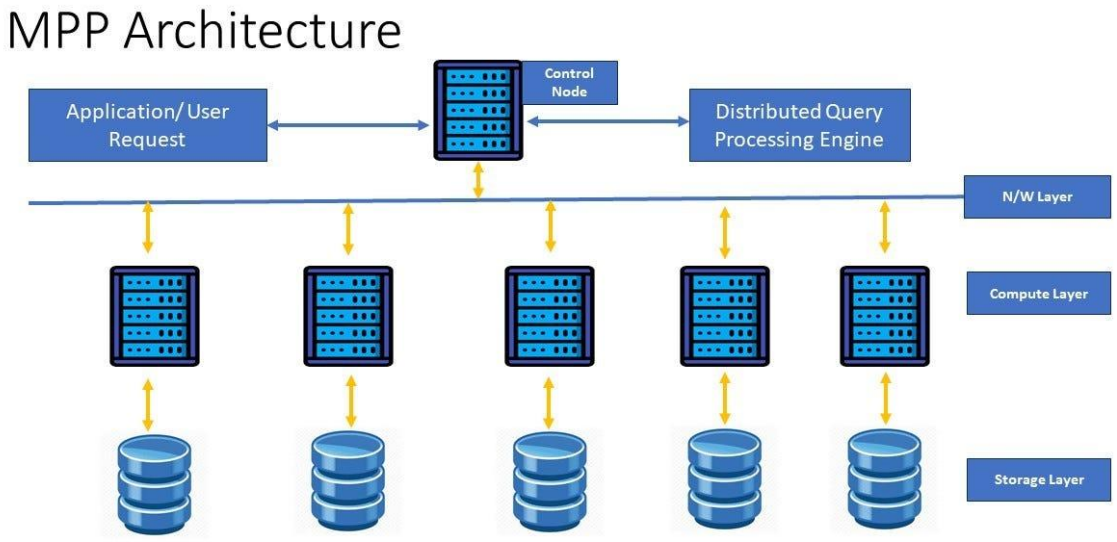
Query Processing and Optimization

Massively Parallel Processing (MPP):

- Distributes data and query load across multiple nodes for efficient processing.

Query Optimization:

- Redshift's query optimizer evaluates multiple execution strategies and selects the most efficient one.



Redshift Deployment Options

Provisioned Clusters

With provisioned clusters, you manage your Amazon Redshift resources manually. This option gives you complete control over cluster sizing, scaling, and maintenance windows.

Redshift Serverless

Amazon Redshift Serverless lets you access and analyze data without the configurations of a provisioned data warehouse. Resources are automatically provisioned, and capacity is intelligently scaled to deliver fast performance for demanding workloads.

Key benefits of Redshift Serverless:

1. **Automatic resource provisioning**: No need to manually configure cluster size
2. **Pay-per-use pricing**: You don't incur charges when the data warehouse is idle
3. **Auto-scaling**: Capacity adjusts based on workload demands
4. **Simplified management**: No need to monitor or tune the cluster

Redshift Serverless organizes resources using namespaces (for database objects) and workgroups (for compute resources). Workgroups house compute resources like Redshift Processing Units (RPUs), security groups, and usage limits.

Setting Up AWS Redshift

Setting Up a Provisioned Cluster

1. **Sign in to AWS Console**: Navigate to the Redshift service
2. **Create Cluster**: Choose "Create cluster" and configure settings:
    1. Cluster identifier
    2. Node type
    3. Number of nodes
    4. Database configurations (name, port, credentials)
    5. Network and security settings
    6. Maintenance and backup options
3. **Launch Cluster**: Review settings and create the cluster

Setting Up Redshift Serverless

1. **Sign in to AWS Console**: Navigate to the Redshift service
2. **Create Serverless Workspace**: Configure settings:
    1. Namespace name (for database objects)
    2. Workgroup name (for compute resources)
    3. Network and security settings
    4. RPU capacity settings (optional)
3. **Create Workspace**: Review settings and create the serverless workspace

Use Cases

Data Warehousing:

- Centralized repository for structured and semi-structured data.

Business Intelligence:

- Supports integration with BI tools like Tableau, Looker, and Amazon QuickSight.

Machine Learning:

- Provides data for training and inference in machine learning models.

Redshift ML

Amazon Redshift ML makes it easy for data analysts and database developers to create, train, and apply machine learning models using familiar SQL commands. It leverages Amazon SageMaker without requiring users to learn new tools or languages.

- Example of creating a model:

```sql
CREATE MODEL customer_churn_model
FROM (SELECT * FROM customer_data)
TARGET churn_indicator
FUNCTION predict_churn
IAM_ROLE 'arn:aws:iam::123456789012:role/RedshiftML'
SETTINGS (
    S3_BUCKET 'redshift-ml-models'
);
```

Using the model:

```sql
SELECT customer_id, predict_churn(age, income, usage_metrics) AS churn_prediction
FROM customer_current
WHERE predict_churn(age, income, usage_metrics) = 1;
```

amazon EMR

Introduction to Amazon EMR

What is Amazon EMR?

- Amazon EMR (Elastic MapReduce) is a managed cloud-native big data platform provided by Amazon Web Services (AWS).
- It simplifies running big data frameworks, such as Apache Hadoop and Apache Spark, on AWS to process and analyze vast amounts of data.

Key Features:

- **Scalability:** Handles large-scale data sets, allowing dynamic scaling of resources based on workload demands.
- **Performance:** Utilizes distributed processing frameworks to decrease command execution time.
- **Simplified Management**: AWS handles provisioning, configuration, and tuning of clusters
- **Flexibility**: Supports multiple deployment options and various open-source big data frameworks
- **Integration:** Seamlessly integrates with various AWS services like Amazon S3, Amazon RDS, and Amazon DynamoDB.
- **Security**: Integrates with AWS security services for comprehensive protection

Use Cases

- Big data processing and analytics
- Machine learning
- Extract, Transform, Load (ETL) operations
- Log analysis
- Real-time streaming
- Data transformations
- Clickstream analysis
- Genomics processing
- Predictive analytics

EMR Architecture Overview

Core Components

AWS EMR architecture consists of several layers, each providing specific functionality to the cluster:

1. Storage Layer

1. Amazon S3 (primary data store)
2. HDFS (Hadoop Distributed File System)
3. EMRFS (EMR File System)
4. Local instance storage

2. Cluster Layer

1. **Primary Node (formerly Master Node)**: Manages the cluster and coordinates distribution of data and tasks
2. **Core Nodes**: Run tasks and store data in HDFS
3. **Task Nodes**: Run tasks but don't store data

3. Processing Layer

1. Open-source frameworks like Hadoop, Spark, Hive, and others
2. EMR-optimized versions of these frameworks

Core Components of Amazon EMR

1. Clusters:

- The central component of Amazon EMR, composed of Amazon EC2 instances. Each instance in the cluster is called a node.

2. Node Types:

- **Primary Node:** Manages the cluster by running software components to coordinate the distribution of data and tasks among other nodes for processing.
- **Core Node:** Runs tasks and stores data in the Hadoop Distributed File System (HDFS) on your cluster.
- **Task Node:** Runs tasks but does not store data in HDFS. Task nodes are optional.

Data Storage and Processing

Storage Options:

- **Amazon S3:** EMR File System (EMRFS) allows EMR clusters to directly access data stored in Amazon S3 as if it were a file system like HDFS.
- **HDFS:** A distributed and scalable Hadoop file system enabling data distribution across instances in the cluster.

Processing Frameworks:

- **Apache Hadoop MapReduce:** An open-source programming model for distributed computing.
- **Apache Spark:** A cluster framework and programming model for processing big data workloads, supporting multiple interactive query modules such as SparkSQL

EMR Studio and Notebooks

EMR Studio:

- An integrated development environment (IDE) that makes it easy for data scientists and data engineers to develop, visualize, and debug data engineering and data science applications written in R, Python, Scala, and PySpark.

EMR Notebooks:

- Provides a managed environment based on Jupyter Notebooks for interactive data analysis and visualization.

Deployment Options

AWS EMR offers four distinct deployment options to meet different operational requirements:

1. EMR on EC2

The traditional deployment model where EMR runs on Amazon EC2 instances.

Key features:

- Full control over cluster configuration and instance selection
- Customizable AMIs (Amazon Machine Images)
- Ability to install custom software
- Ideal for workloads requiring specific hardware configurations

Use when:

- You need maximum control and flexibility
- You want to customize underlying infrastructure
- You need specific EC2 instance types for performance optimization

Deployment Options

A serverless deployment option that eliminates the need to configure, manage, or scale clusters.

Key features:

- Automatic resource provisioning and scaling
- Pay only for resources used by your applications
- Zero infrastructure management
- Currently supports Apache Spark and Apache Hive

Use when:

- You want to simplify operations
- You have variable or unpredictable workloads
- You want to avoid managing clusters
- You prefer focusing on applications rather than infrastructure

Setting Up AWS EMR

Setting Up EMR on EC2

1. **Sign in to AWS Console**: Navigate to the EMR service
2. **Create Cluster**: Choose "Create cluster" and configure:
    1. Cluster name
    2. Release version (applications and versions)
    3. Instance types and count
    4. Networking settings
    5. Security configurations
    6. Bootstrap actions (optional)
3. **Launch Cluster**: Review settings and create

Setting Up EMR Serverless

1. **Sign in to AWS Console**: Navigate to the EMR service
2. **Create Application**:
    1. Choose application type (Spark or Hive)
    2. Select release version
    3. Configure capacity settings (optional)
    4. Set up network and security options
3. **Submit Jobs**: Use the EMR Studio, AWS CLI, or SDK to submit jobs

Data Processing Frameworks

EMR supports a variety of open-source frameworks for data processing. The latest EMR releases include:

Apache Hadoop

The foundation framework providing:

- HDFS (Hadoop Distributed File System)
- YARN (resource management)
- MapReduce (programming model)

Apache Spark

A unified analytics engine offering:

- In-memory processing
- Support for SQL, streaming, machine learning, and graph processing
- Performance up to 100x faster than traditional MapReduce

Apache Hive

Data warehouse software providing:

- SQL-like interface (HiveQL)
- Structured data querying capabilities
- Integration with various storage system

Data Processing Frameworks

Presto and Trino

Distributed SQL query engines for:

- High-performance interactive analytics
- Querying data across multiple sources

Apache Flink

A stream processing framework offering:

- Real-time data processing
- Event-time processing
- State management

HBase

A NoSQL database providing:

- Random, real-time read/write access
- Storage for large tables with billions of rows

Data Processing Frameworks

Other Supported Frameworks

- Livy: REST interface for Spark
- Hudi: Data lake storage framework
- Delta Lake: Storage layer for reliability
- Iceberg: Table format for huge analytic datasets
- TensorFlow: Machine learning framework

Framework Selection Guidance

When choosing frameworks, consider:

- Data processing needs (batch vs. streaming)
- Query complexity and performance requirements
- Skill levels and team expertise
- Integration requirements with other systems
- Operational considerations

Working with Data in EMR

Data Storage Options

1.Amazon S3:

1. Recommended for persistent, long-term storage
2. Accessible from multiple clusters
3. Separation of compute and storage
4. Integrated with EMRFS for optimized access

2.HDFS (Hadoop Distributed File System):

1. Fast access for temporary data
2. Data is lost when cluster terminates
3. Best for intermediate processing results

3.Amazon DynamoDB:

1. For NoSQL database integration
2. Direct access from EMR applications

Working with Data in EMR

Data Input and Output

1.Loading Data to EMR:

1. Direct read from S3 using EMRFS
2. Import from DynamoDB
3. Streaming data from Kinesis
4. Loading from JDBC sources

2.Exporting Processed Data:

1. Writing to S3
2. Exporting to DynamoDB
3. Saving to other AWS data stores

Working with Data in EMR

Data Formats

EMR supports multiple data formats:

- Parquet (columnar format, optimized for performance)
- ORC (Optimized Row Columnar)
- Avro (row-based format with schema)
- JSON
- CSV
- Text

Open Table Formats

EMR 7.x and 6.x releases include support for:

- **Apache Hudi**: For data lake transactions and change data capture
- **Apache Iceberg**: For table format with schema evolution and partition evolution
- **Delta Lake**: For ACID transactions and time travel capabilities

Data Processing Patterns

1. **Batch Processing**:
    1. Processing large volumes of data periodically
    2. ETL jobs for data warehousing
2. **Interactive Analysis**:
    1. SQL queries with Hive, Presto, or Spark SQL
    2. Ad-hoc analysis in notebooks
3. **Streaming Processing**:
    1. Real-time data processing with Spark Streaming or Flink
    2. Processing data from Kinesis or Kafka
4. **Machine Learning**:
    1. Building and deploying models with Spark MLlib
    2. Feature engineering and model scoring

Data Governance

1. **AWS Glue Data Catalog**:
    1. Centralized metadata repository
    2. Table and database definitions
    3. Schema management
2. **AWS Lake Formation**:
    1. Centralized permissions management
    2. Data discovery and cataloging
    3. Secure data sharing

Auditing and Monitoring

1. **CloudTrail Integration**:
    1. Audit logging for API calls
    2. User activity tracking
2. **Security Monitoring**:
    1. Integration with Amazon GuardDuty
    2. Security findings for potential issues

Monitoring Tools

1. **CloudWatch Integration**:
    1. Metric collection and visualization
    2. Alarm configuration
    3. Log aggregation and analysis
2. **Ganglia**:
    1. Cluster performance monitoring
    2. Resource utilization tracking
3. **Prometheus and Grafana**:
    1. Custom monitoring solutions
    2. Advanced visualization

Critical Metrics to Monitor

1. **Cluster Health**:
    1. Node status
    2. Service availability
    3. HDFS health
2. **Resource Utilization**:
    1. CPU usage
    2. Memory consumption
    3. Disk space
    4. Network I/O
3. **Application Metrics**:
    1. Job completion rates
    2. Processing latency
    3. Error rates
    4. Throughput

Cost Management

EMR Pricing Models

1.On-Demand Instances:

1. Pay per hour with no long-term commitments
2. Higher rate but maximum flexibility

2.Reserved Instances:

1. Lower hourly rate with 1 or 3-year commitments
2. Significant discount for predictable workloads

3.Spot Instances:

1. Use unused EC2 capacity at steep discounts
2. Subject to interruption based on capacity availability

4.EMR Serverless Pricing:

1. Pay only for vCPU, memory, and storage resources used
2. No need to pay for idle capacity

Cost Optimization Strategies

1. **Cluster Management**:
    1. Use transient clusters for batch processing
    2. Leverage auto-termination for idle clusters
    3. Implement managed scaling
2. **Instance Strategy**:
    1. Mix instance types (On-Demand, Spot)
    2. Use Instance Fleets for flexible allocation
    3. Leverage Graviton processors for better price-performance
3. **Storage Optimization**:
    1. Use S3 for persistent storage
    2. Compress data appropriately
    3. Implement lifecycle policies for old data
4. **Workload Optimization**:
    1. Improve job efficiency to reduce runtime
    2. Schedule jobs during off-peak hours
    3. Batch small processing jobs

Cost Monitoring

1.AWS Cost Explorer:

1. Track EMR spending by cluster, user, or project
2. Analyze cost trends
3. Create budget alerts

2.AWS Budgets:

1. Set spending limits
2. Configure alerts for potential overruns

3.Cost Allocation Tags:

1. Tag resources for cost attribution
2. Track costs by department, project, or environment

Integration with AWS Ecosystem

Data Integration

1.Storage Services:

1. Amazon S3: Primary data lake storage
2. Amazon EFS: Shared file system
3. Amazon FSx for Lustre: High-performance file system

2.Database Services:

1. Amazon RDS: Relational databases
2. Amazon DynamoDB: NoSQL database
3. Amazon Redshift: Data warehouse

3.Streaming Services:

1. Amazon Kinesis: Real-time streaming data
2. Amazon MSK: Managed Kafka service

Analytics Integration

1.Amazon Athena:

1. Serverless query service for S3 data
2. Shares metadata with EMR through Glue Data Catalog

2.Amazon QuickSight:

1. Business intelligence and visualization
2. Connect to EMR data sources

3.Amazon SageMaker:

1. Machine learning platform
2. Use EMR for data preparation

Data Governance Integration

1.AWS Glue:

1. Metadata catalog shared with EMR
2. ETL job orchestration

2.AWS Lake Formation:

1. Fine-grained access control for data lakes
2. Integration with EMR for permission enforcement

3.Amazon DataZone:

1. Data discovery and governance
2. Metadata management

Operational Integration

1. **AWS CloudFormation**:
    1. Infrastructure as code for EMR
    2. Automated cluster provisioning
2. **AWS Step Functions**:
    1. Workflow orchestration
    2. EMR job coordination
3. **Amazon EventBridge**:
    1. Event-driven EMR management
    2. Automated responses to cluster events

Integration Architecture Patterns

1.Data Lake Architecture:

1. S3 as central storage
2. EMR for processing
3. Athena and QuickSight for analysis

2.Streaming Analytics:

1. Kinesis or MSK for data ingestion
2. EMR for stream processing
3. DynamoDB or S3 for processed data

3.Machine Learning Pipeline:

1. EMR for data preparation
2. Feature Store for feature management
3. SageMaker for model training and deployment

Hands-On Exercises

Exercise 1: Setting Up an EMR Cluster

1. Launch a basic EMR cluster using the AWS Management Console
2. Connect to the cluster using SSH
3. Explore the cluster web interfaces (YARN, Spark, etc.)
4. Submit a simple job and monitor its execution

Exercise 2: Processing Data with Spark

1. Prepare sample dataset in S3
2. Create a Spark application to process the data
3. Submit the application to EMR
4. Analyze the results and optimize the application

Exercise 3: EMR Serverless Deployment

1. Create an EMR Serverless application
2. Configure pre-initialized capacity
3. Submit Spark or Hive jobs
4. Monitor job execution and resource usage

Exercise 4: Security Implementation

1. Create a security configuration with encryption
2. Set up Kerberos authentication
3. Implement Lake Formation integration
4. Test fine-grained access controls

Exercise 5: Performance Optimization

1. Benchmark a standard EMR configuration
2. Apply performance optimization techniques
3. Measure the impact of optimizations
4. Develop a performance tuning checklist

Lab: Processing Logs by Using Amazon EMR

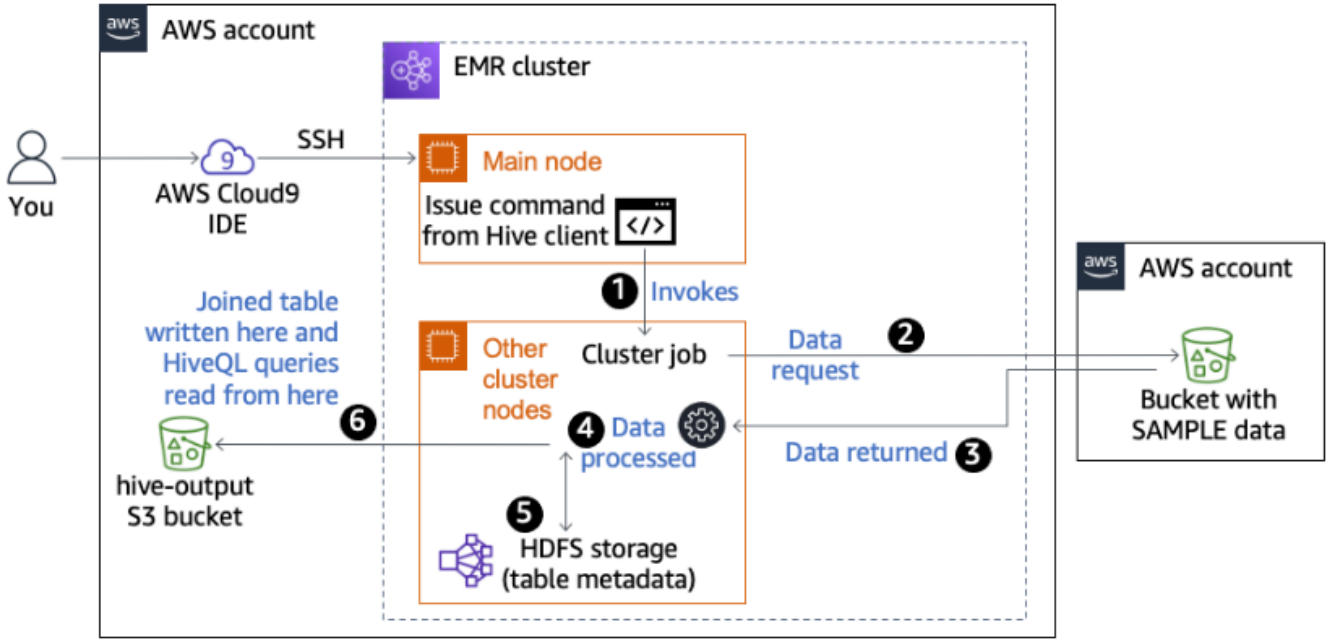
In this lab, you will use Amazon EMR to process logs that are stored in an S3 bucket.

To process the data in Amazon EMR, you will use Apache Hive.

Hive is an open-source data warehouse infrastructure framework that is built on top of Hadoop and is often used to process large datasets.

With Hive, you can query large datasets with SQL-like queries.

When you run a Hive query, the request is translated into a Yet Another Resource Negotiator (YARN) job. YARN is the part of the Hadoop software that coordinates job scheduling.

After completing this lab, you should be able to do the following:

- Launch an EMR cluster through the AWS Management Console.
- Run Hive interactively.
- Use Hive commands to create tables from log data that is stored in Amazon S3.
- Use Hive commands to join tables and store the joined table in Amazon S3.
- Use Hive to query tables that are stored in Amazon S3.